

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

*Student Assessment Document*

### FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.8 VECTORS, FINAL EXPLANATION

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Driving question: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations, and what mathematical tool do we need to describe a journey completely? Answer every section in the space under each prompt. Use what you learned in Lessons 1 to 8: the Kencom matatu map, the Nairobi grid with Kencom Bus Stop as the origin O, and the Nakuru journey map. In the calculation sections, show each step of your working and write the correct units (take 1 grid unit = 1 km). Use the terms scalar, vector, distance and displacement carefully. Where a question asks you to explain, write full sentences and connect your answer back to the two matatus from Kencom.

### Part 1: Why is '8 km' not enough?

Two matatu drivers both travel 8 km from Kencom Bus Stop in Nairobi's CBD. One arrives at Westlands (north-west on the map) and the other arrives at South B (south-east on the map). (a) Is '8 km' a scalar or a vector quantity? Explain what it tells you and what it does not tell you. (b) Give one example of a scalar and one example of a vector from everyday Kenyan life, and say why each belongs in its group. (c) Use the words magnitude and direction to explain why the two matatus end up in different places.

*Write your answer here:*

## Part 2: Writing journeys as column vectors

On the simplified Nairobi coordinate grid used in Lessons 5 and 6, Kencom Bus Stop is the origin O. On this grid, one Westlands-route journey is the column vector  $w = \begin{pmatrix} -6 \\ 5 \end{pmatrix}$  and one South B-route journey is the column vector  $s = \begin{pmatrix} 4 \\ -6 \end{pmatrix}$ . (a) Explain what the top number and the bottom number in a column vector tell you. (b) Calculate the magnitude of  $w$  and the magnitude of  $s$ , correct to 1 decimal place. (c) Compare your two magnitudes with the '8 km' labels on the map. (d) Explain why the two column vectors are different even though their magnitudes are similar.

*Write your answer here:*

### Part 3: Position vectors, displacement and midpoint

On the same grid, KICC is at point A and Uhuru Park is at point B. The position vectors are  $OA = (2; 3)$  and  $OB = (-1; 2)$ . (a) Find the displacement vector AB. (b) Find the magnitude of AB, correct to 1 decimal place. (c) A matatu travels from A to B. Find the coordinates of the midpoint of this journey. (d) A translation moves every point of an object by the same vector. State the translation vector that moves point A to point B and explain how you found it.

*Write your answer here:*

#### Part 4: Adding, scaling and reversing journeys

(a) A bodaboda rider goes from O to A, then from A to B, using the position vectors from Part 3. Write the two legs as column vectors, add them to find the resultant, and say what the resultant represents on the grid. (b) Explain in words how the triangle law (tip-to-tail) would show this resultant on squared paper. (c) The displacement vector  $d = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ . Write  $3d$  and  $-d$  as column vectors and describe how each compares with  $d$  in magnitude and direction. (d) Write the journey from B back to O as a vector and explain how it is related to OB.

*Write your answer here:*

## Part 5: Final explanation and the Nakuru journey

Write a full answer to the driving question: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations, and what mathematical tool do we need to describe a journey completely? In your answer: (a) use the terms scalar, vector, magnitude and direction. (b) Refer to your column vector results from Part 2. (c) Explain how you would use vectors to describe a three-leg journey in Nakuru from the CBD Terminus to Westside Mall, then the ASK Showground, then the Lake Nakuru Viewpoint. Say how the total distance travelled differs from the displacement from start to finish. (d) Describe how your thinking has changed since Day 1.

*Write your answer here:*

| RUBRIC                                                |                                                                                                                                                                     |                                                                                                                              |                                                                                                        |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Criterion                                             | Excellent (4)                                                                                                                                                       | Proficient (3)                                                                                                               | Developing (1–2)                                                                                       |
| Scalar and vector concepts                            | Correctly identifies 8 km as a scalar, gives accurate Kenyan examples of both types, and clearly explains that magnitude and direction together describe a journey. | Identifies scalar and vector correctly with mostly accurate examples, but the explanation of why direction matters is brief. | Confuses scalar and vector, or gives examples without a reason.                                        |
| Column vector and magnitude calculations              | Reads components correctly and applies $\sqrt{x^2 + y^2}$ accurately to get 7.8 km and 7.2 km, with units and an honest comparison with the 8 km labels.            | Calculates the magnitudes with a minor slip in rounding or units, and makes a limited comparison with the labels.            | Cannot interpret components or uses an incorrect formula for magnitude.                                |
| Position vectors, resultant, midpoint and translation | Correctly finds $AB = (-3; -1)$ , its magnitude, the midpoint $(0.5; 2.5)$ , the resultant $(-1; 2)$ , the scaled vectors, and                                      | Gets most of the vector operations right with one or two arithmetic or sign errors, and links ideas partially.               | Makes frequent errors in subtraction, addition or scalar multiplication and does not link the results. |

|                                           |                                                                                                                                                                                                 |                                                                                                            |                                                                                           |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                                           | links translation to displacement.                                                                                                                                                              |                                                                                                            |                                                                                           |
| Final explanation of the driving question | Gives a coherent explanation using the vocabulary, the Part 2 evidence, and the multi-leg Nakuru journey, distinguishes distance from displacement, and reflects on how their thinking changed. | Answers the driving question with correct vocabulary but gives limited evidence or only a weak reflection. | Gives an incomplete or unclear answer that does not use the vector ideas or the evidence. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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